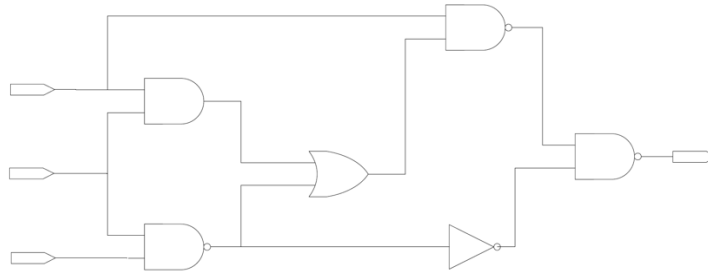
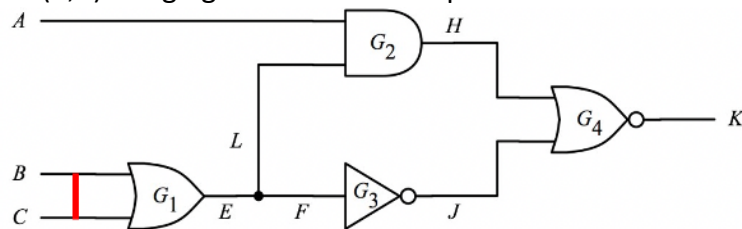


Homework # 2

1. For the logic circuit in figure 1, Compute a) the total number of single stuck-at faults b) the total number of stuck-open faults c) the total number of all possible multiple stuck-at fault combinations (assume 2-inputs NANDs and NORs use 4 Transistors, 2-input ANDs and ORs use 6 transistors, and NOTs use 2 transistors).



2. Fill in the table with (B,C) bridging fault. Which test pattern detects a wired-OR fault?



Inputs A B C	Fault-free Output	Wired- OR	Wired- AND	B- Dominant
0 0 0				
0 0 1				
0 1 0				
0 1 1				
1 0 0				
1 0 1				
1 1 0				
1 1 1				

3. Fill in the table with a Wired-AND feedback bridging fault (x, z).

Inputs X Y	Fault-free Output	Faulty Output
0 0	0	
0 1	1	
1 0	1	
1 1	1	

